

Jalina Hirushan

Final-Year IT Undergraduate / AI/ML Engineering

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Profile

Final-year Information Technology undergraduate at the University of Moratuwa with Software Engineering internship experience and a focus on AI/ML engineering. Hands-on experience building LLM-powered applications, RAG and Graph-RAG systems, semantic retrieval pipelines, agentic workflows, and full-stack software. Interested in applying strong software engineering foundations to practical AI systems and intelligent developer tooling.

Education

BSc (Hons) in Information Technology — *University of Moratuwa* 2023–2027
Final-Year Undergraduate CGPA: 3.35 (up to Semester 5)

GCE Advanced Level — *Nalanda College* 2021
Combined Mathematics – A, Chemistry – A, Physics – B Z-score: 1.758

Technical Skills

- **Programming:** Python, JavaScript, TypeScript, Dart
- **AI & LLM Engineering:** LLM Applications, RAG, Graph-RAG, Vector Embeddings, Semantic Search, Knowledge Graphs, Agentic Workflows, Prompt Engineering, Structured LLM Outputs
- **Machine Learning:** scikit-learn, XGBoost, Classification, Feature Engineering, Cross-Validation, Model Evaluation
- **AI Platforms & Tools:** OpenAI API, Google Gemini API, OpenRouter, Neo4j
- **Web & Backend:** React, Next.js, Node.js, .NET, REST APIs
- **Cloud & DevOps:** AWS, Docker, Terraform, CI/CD, Git, GitHub Actions
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite, Supabase

Work Experience

Software Engineering Intern — **Infinity Innovators (Pvt.) Ltd.** Jan 2026 – Jul 2026

- Developed an AI-powered RAG chatbot and intelligent search functionality for a marketplace platform using vector embeddings and LLM APIs to support contextual product discovery.
- Contributed to full-stack production systems using React, TypeScript, Node.js and AWS, including multi-tenant dealer platform functionality and dynamic business-data integration.
- Worked with AWS infrastructure and Terraform/IaC, investigated CI/CD failures, contributed infrastructure configurations, and supported automated deployment workflows.
- Participated in Agile development, Git-based collaboration, pull requests, code reviews, debugging, testing, and production-oriented software delivery.

Freelance Web & Mobile Developer — **Upwork** 2025 – Present

- Delivered 5+ mobile applications for clients using Flutter, based on custom requirements.

Projects

TerraFix — **AI-Powered Terraform Failure Diagnosis & Verified Repair** 2026
Ascentic AI Launch Pad 2026 — Top 20 from 700+ applicants

- Built an agentic system that investigates semantic Terraform CI failures using bounded repository context and LLM reasoning, producing structured root-cause diagnoses and source-level repair proposals.
- Designed a deterministic repair pipeline that converts structured AI edits into Git patches and validates proposed changes using Terraform tooling before allowing human-controlled application to a pull request.
- Implemented bounded multi-stage reasoning, model routing, failure classification, GitHub integration, verification controls, and observability for model calls, tokens, latency and cost.
- **Technologies:** Python, LLMs, OpenRouter, Terraform, Next.js, TypeScript, PostgreSQL, Docker, GitHub Apps

Physics Knowledge Graph & Graph-RAG System

2026

Multilingual knowledge retrieval and question-answering system

- Built a structured physics knowledge base containing concepts, definitions, formulas, practical applications and visual knowledge, represented as a Neo4j knowledge graph.
- Developed a Graph-RAG retrieval pipeline using multilingual embeddings, semantic retrieval, graph relationships and query translation to answer Sinhala physics questions using relevant knowledge context.
- Built an evaluation workflow for generated answers and experimented with graph expansion, cross-topic relationships and retrieval strategies.
- **Technologies:** Python, Neo4j, Graph-RAG, multilingual-e5, Vector Embeddings, LLMs, Semantic Retrieval

Synapse — AI-Powered Second Brain

2026

Semantic note retrieval and contextual AI chat platform

- Developed a full-stack knowledge application with semantic note retrieval using vector embeddings and contextual AI chat powered by Google Gemini APIs.
- Implemented secure authentication, protected routes, persistent data storage and tier-based usage management.
- **Technologies:** Next.js, React, TypeScript, Supabase, PostgreSQL, Vector Embeddings, Google Gemini API, Docker
- **Demo:** synapse-opal-five.vercel.app

Achievements

- **Ascentic AI Launch Pad 2026** — Selected to the **Top 20 from 700+** applicants
- **HackMoral 7.0** — Runners-up
- **CryptX 1.0** — Finalist
- **CodeRally 6.0** — Advanced Tier Finalist
- **UGRAM Award Winner** — Inspiring Rotaract Projects of South Asia

Leadership & Activities

- **Project Co-chairperson** — **Project Aya, Rotaract Club of University of Moratuwa:** Contributed to Sinhala language data collection for Cohere's multilingual Aya initiative.
- **Batch Representative** — **Faculty of Information Technology, University of Moratuwa**
- **Open-Source Contributor** — **Hacktoberfest 2024 & 2025**

References

Dr. S. C. Premaratne

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